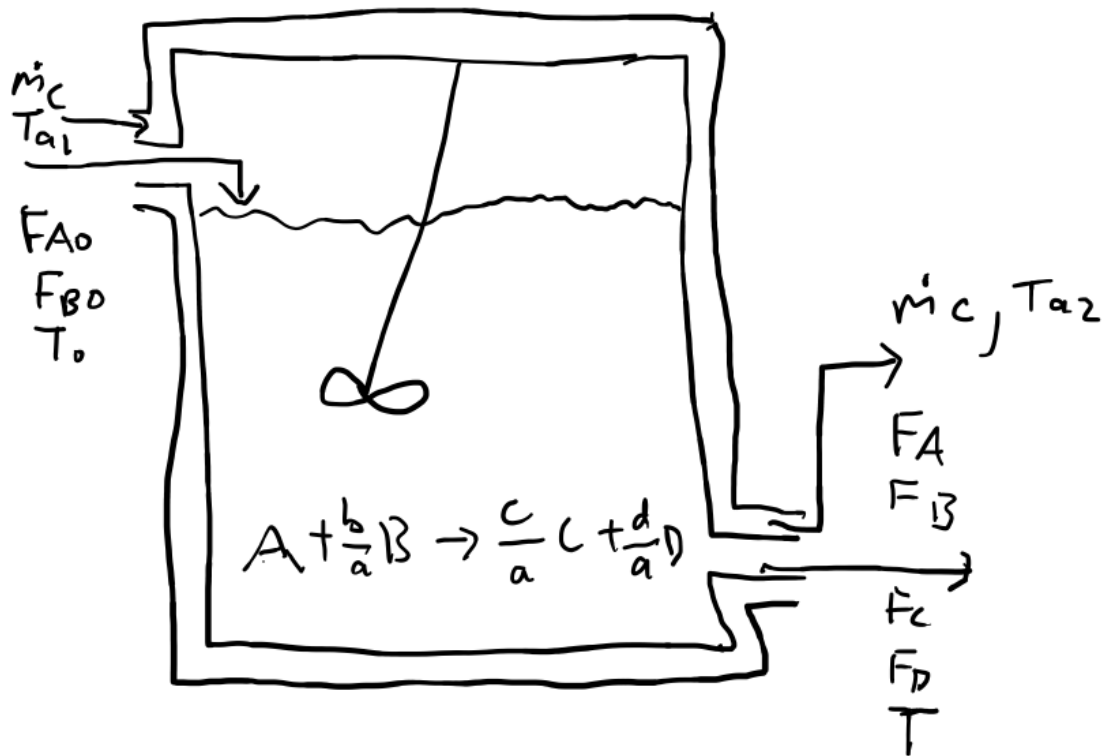


18 Energy Balances for CSTRs

Let's derive the energy balance for a CSTR:



Starting from the general form (assuming steady state):

$$0 = \dot{Q} + \dot{W}_s + \sum F_{i0} H_{i0} - \sum F_i H_i$$

In the CSTR, the stirrer is doing work **on** the reactor fluid and hence increasing its energy. However, this amount of work is usually very small relative to the heat of reaction. So, we can usually set $\dot{W}_s = 0$. Exception: If the fluid is viscous and you are given a value for $\dot{W}_s$.

$\dot{Q}$ term: Two cases:

- $T_{a1} \neq T_{a2}$: Rate of heat transfer varies over the heat transfer area. In this case, use *log mean temperature difference*

$$\dot{Q} = \frac{UA(T_{a1} - T_{a2})}{\ln \left[\frac{(T - T_{a1})}{(T - T_{a2})} \right]}$$

- $T_{a1} = T_{a2}$: Rate of heat transfer is constant over the heat transfer area. In this case, use

$$\dot{Q} = UA(T_a - T)$$

For the purposes of this derivation, let's use the second case. Then:

$$0 = UA(T_a - T) + F_{A0}H_{A0} + F_{B0}H_{B0} - F_A H_A - F_B H_B - F_C H_C - F_D H_D$$

In the case of just one reaction, F terms can be related to F_0 terms using X :

$$F_A = F_{A0}(1 - X) \quad F_C = \frac{c}{a} F_{A0} X$$

$$F_B = F_{A0} \left(\theta_B - \frac{b}{a} X \right) \quad F_D = \frac{d}{a} F_{A0} X$$

Expanding terms:

$$0 = UA(T_a - T) + F_{A0}H_{A0} + F_{B0}H_{B0} - F_{A0}(1 - X)H_A - F_{A0} \left(\theta_B - \frac{b}{a} X \right) H_B - \frac{c}{a} F_{A0} X H_C - \frac{d}{a} F_{A0} X H_D$$

Further simplify by using the definition that $\theta_i = \frac{F_{i0}}{F_{A0}}$:

$$0 = UA(T_a - T) + F_{A0}H_{A0} + F_{B0}H_{B0} - F_{A0}H_A + F_{A0}H_A X - F_{A0}\theta_B H_B + \frac{b}{a} F_{A0}H_B X - \frac{c}{a} F_{A0}X H_C - \frac{d}{a} F_{A0}X H_D$$

Grouping like terms:

$$0 = U_A (T_a - T) + F_{A0} (H_{A0} - H_A) + F_{B0} \Theta_B (H_{B0} - H_B) + F_{A0} X (H_A + \frac{b}{a} H_B - \frac{c}{a} H_C - \frac{d}{a} H_D)$$

Where $\Delta H^{rxn} = \frac{d}{a} H_D + \frac{c}{a} H_C - \frac{b}{a} H_B - H_A$. Therefore:

$$0 = U_A (T_a - T) + F_{A0} (H_{A0} - H_A) + F_{B0} \Theta_B (H_{B0} - H_B) + F_{A0} X (-\Delta H^{rxn})$$

$H_{i0} - H_i$ terms are **sensible heats**, equal to $C_p(T_0 - T)$. Substituting in:

$$0 = U_A (T_a - T) + F_{A0} C_{PA} (T_0 - T) + F_{B0} \Theta_B C_{PB} (T_0 - T) + F_{A0} X (-\Delta H^{rxn})$$

Solving for X:

$$X_{EB} = \frac{F_{A0} C_{PA} (T_0 - T) + F_{B0} C_{PB} \Theta_B (T_0 - T) + U_A (T_a - T)}{F_{A0} (\Delta H^{rxn})}$$

Ultimately, we have 2 unknowns, X and T, hence, we need 2 equations: a mole balance and an energy balance. We denote this equation EB to denote that it is the energy balance.

The mole balance is: $V = \frac{F_{A0} X_{MB}}{-r_A}$, where $-r_A = k C_A^{\frac{b}{a}}$, $k = k_1(T_1) \exp \left[\frac{E}{R} \left(\frac{1}{T_1} - \frac{1}{T} \right) \right]$, and $C_i = \frac{C_{A0} \left(\Theta_i - \frac{v_{LR}}{v_{LR}} X \right) T_0}{1 + \epsilon X} \frac{T_0}{T}$.

The mole balance can be solved for X, giving a function $X_{MB}(T)$.

We now have a system of 2 equations and 2 unknowns, which can be solved by hand, with mathematical software, or graphically.